
Mi Tesis en 180 Segundos — ICI2ST 2026

Vision Transformers for Breast Cancer Prediction in Structural Mammography and Functional Infrared Thermography

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Problem Context

- **Breast cancer** is a leading cause of cancer death in women, in Ecuador and worldwide.
- **Global incidence** is expected to exceed 6 million new cases by 2050.
- **Early detection saves lives**, but reading images is slow and prone to missing subtle findings.

6M+

new annual cases projected worldwide
by 2050

2nd

leading cause of cancer death in
women worldwide

1 in 20

approximate lifetime risk of breast
cancer for women

Research Goal and Proposal

Hypothesis

Under controlled training and evaluation protocols, transformer-based models match or outperform their convolutional baselines for breast-cancer prediction in structural mammography and functional thermography.

Goal

To develop and evaluate transformer-based models for the semantic segmentation of breast lesions in mammography, and for region-of-interest segmentation and malignancy classification of thermal patterns in infrared thermography, using a quantitative, quasi-experimental Design Science Research protocol with patient-level cross-validation.

Research Gap

- **Vision transformers** are little studied for infrared thermography; public datasets rarely pair mammography and thermography for the same patient; and no standardized benchmarking exists, making results hard to compare.

Gap 1

Scarce pectoral-muscle annotations in mammography

Zero-shot MedSAM study.

Gap 2

No per-view thermography benchmark

View-conditioned SegFormer study.

Gap 3

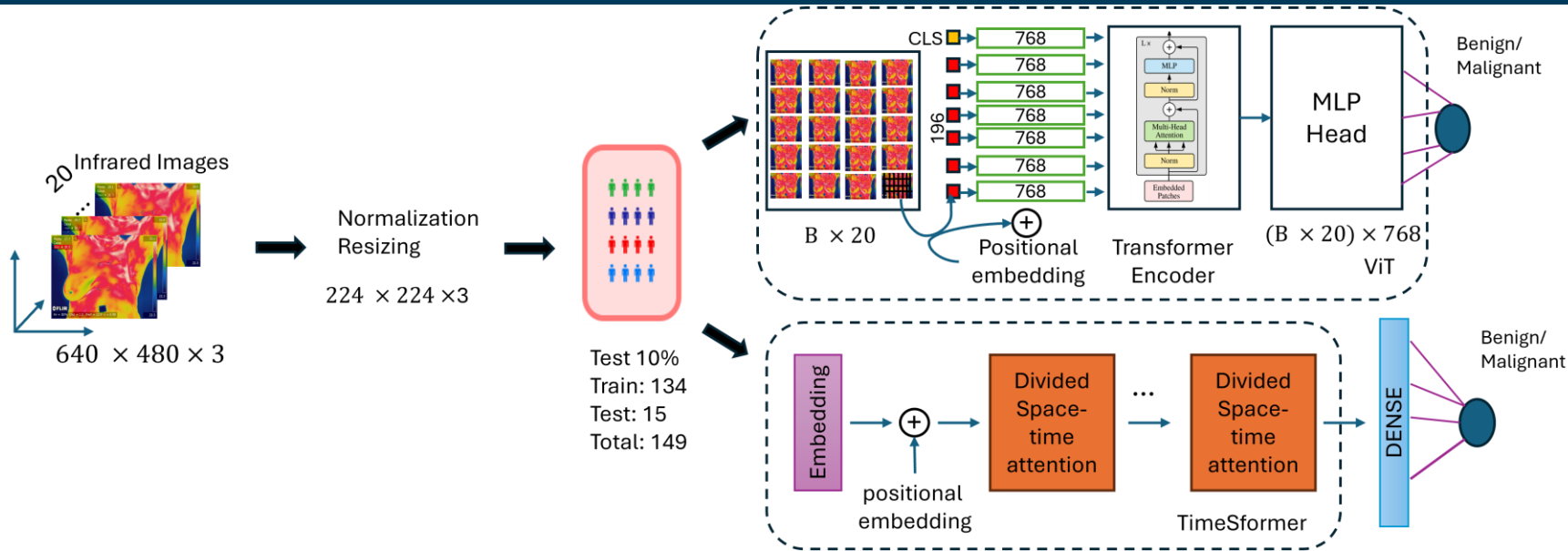
Temporal dynamics underexplored in dynamic infrared

TimeSformer study.

Proposal: *one transformer-based model per gap, trained and evaluated under a common patient-level cross-validation protocol.*

Video Transformers for Dynamic Infrared Thermography

Published · 2024



Study	Model	Protocol	Unit	AUC
Garia et al. (2023)	ViT	Static (SIT)	Image (950)	0.9570
Baffa et al. (2023)	3D-CNN	Dynamic (DIT)	Patient (149)	0.9351
Ours	TimeSformer (ViViT)	Dynamic (DIT)	Patient (149)	0.9444

Vision Transformer pipeline used in this research.

Zero-Shot Pectoral Muscle Segmentation with MedSAM

Accepted · 2025

- **Ground-truth pectoral-muscle masks are scarce.** We tested a zero-shot foundation model: MedSAM prompted by a human expert.
- **On 168 MIAS cases:** mean DSC = 0.865, mean IoU = 0.783, median DSC = 0.909.
- **Accepted for publication** (proceedings pending indexing).

0.865

mean DSC

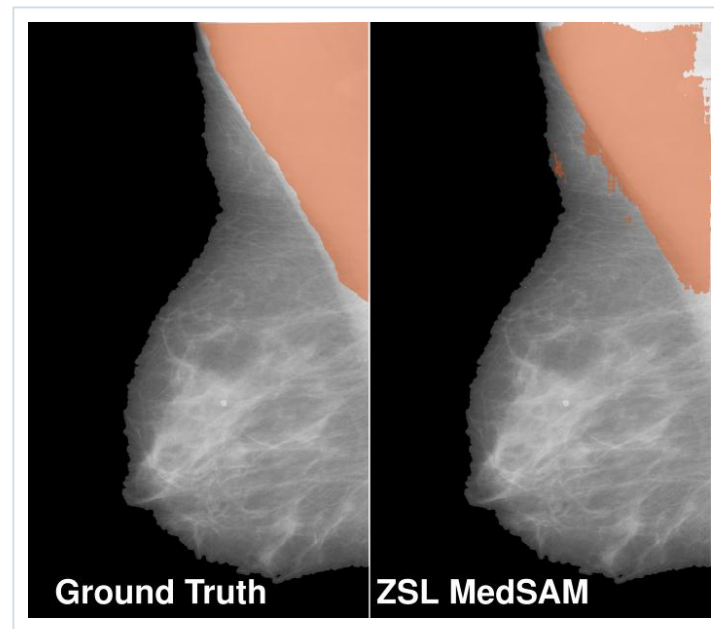
0.783

mean IoU

0.909

median DSC

168 MIAS cases evaluated



Pectoral muscle overlay: ground truth vs zero-shot MedSAM.

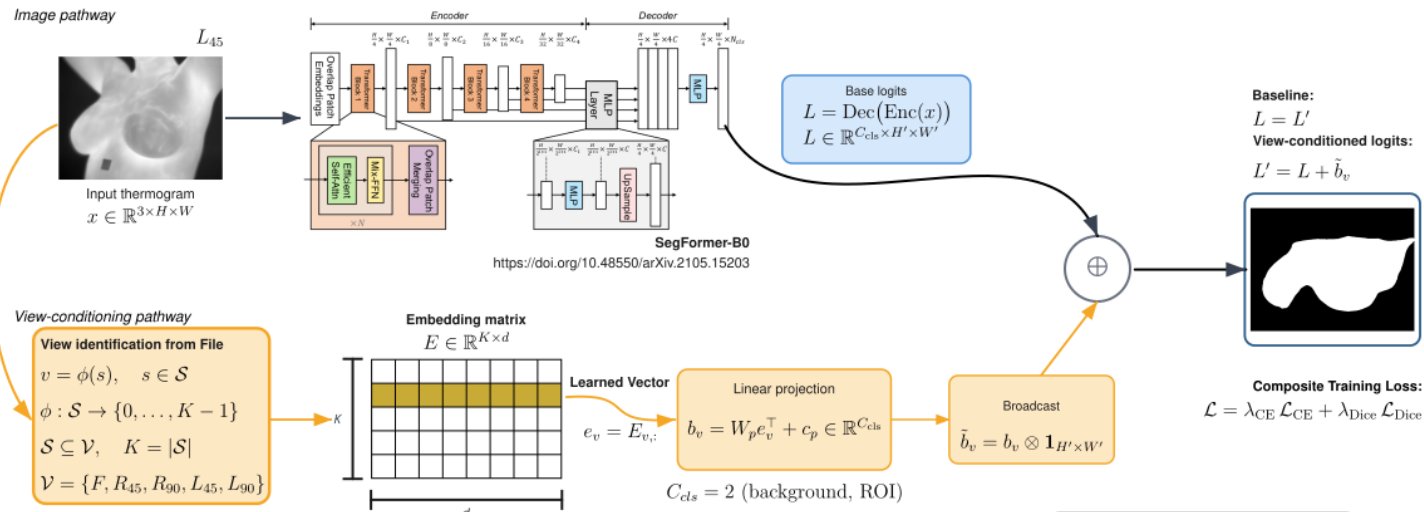
- **One lightweight SegFormer-B0** with a view-conditioning module segments all five canonical views (frontal, oblique 45°, lateral 90°, each side).
- **Improves on both frontal and lateral views** of published DMR–UFF baselines under patient-level 5-fold cross-validation.

Study	Model	View	IoU	DSC
Baffa et al. (2021)	U-Net	Frontal	94.38%	—
Carvalho et al. (2023)	U-Net	Lateral	82.30%	—
Jaincy et al. (2025)	DSU-Net-SCA	Frontal	88.50%	91.70%
Ours	SegFormer-B0	Frontal (5-fold CV)	96.66%	98.29%
Ours	SegFormer-B0	Lateral (5-fold CV)	85.31%	91.86%
Ours	View-conditioned SegFormer-B0	All 5 views (5-fold CV)	92.61%	96.06%

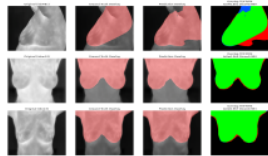
Impact: +2.3 IoU points over the best frontal baseline and +3.0 over the lateral baseline — the first per-view benchmark on DMR–UFF, with a single model for all five views.

View-Conditioned SegFormer-B0 Architecture

View-Conditioned SegFormer-B0: Post-Decoder Logit-Bias Architecture



Qualitative Results:



Quantitative Results:

- Overall: 92.61%
- Frontal: 96.66%
- Lateral: 88.98%

Thank You

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